

Student Name: _____

Class Name: **College Algebra FA26 - 014 MW 2:30**Number of Questions: **25**Instructor Name: **Mamani Velasco, Ruben****Question 1 of 25**

The one-to-one functions g and h are defined as follows.

$$g = \{(-5, 2), (0, -1), (2, 5), (7, 1), (8, 7)\}$$

$$h(x) = \frac{x+3}{7}$$

Find the following.

$$g^{-1}(2) = \underline{\hspace{2cm}}$$

$$h^{-1}(x) = \underline{\hspace{2cm}}$$

$$(h^{-1} \circ h)(4) = \underline{\hspace{2cm}}$$

Question 2 of 25

For each pair of functions f and g below, find $f(g(x))$ and $g(f(x))$.
Then, determine whether f and g are inverses of each other.

Simplify your answers as much as possible.
(Assume that your expressions are defined for all x in the domain of the composition.
You do *not* have to indicate the domain.)

(a) $f(x) = x - 4$ $g(x) = x + 4$ $f(g(x)) =$ $g(f(x)) =$ - f and g are inverses of each other - f and g are <i>not</i> inverses of each other	(b) $f(x) = \frac{1}{3x}$ $g(x) = -\frac{1}{3x}$ $f(g(x)) =$ $g(f(x)) =$ - f and g are inverses of each other - f and g are <i>not</i> inverses of each other
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(a) $f(x) = x - 4$

$g(x) = x + 4$

(b) $f(x) = \frac{1}{3x}, x \neq 0$

$g(x) = -\frac{1}{3x}, x \neq 0$

Find the compositions.

$f(g(x)) =$ _____

$g(f(x)) =$ _____

Find the compositions.

$f(g(x)) =$ _____

$g(f(x)) =$ _____

- Are f and g inverses of each other?
- ☐ f and g are inverses of each other
- ☐ f and g are *not* inverses of each other

- Are f and g inverses of each other?
- ☐ f and g are inverses of each other
- ☐ f and g are *not* inverses of each other

Question 3 of 25

The function h is defined by the following rule.

$$h(x)=\left(\frac{1}{10}\right)^x$$

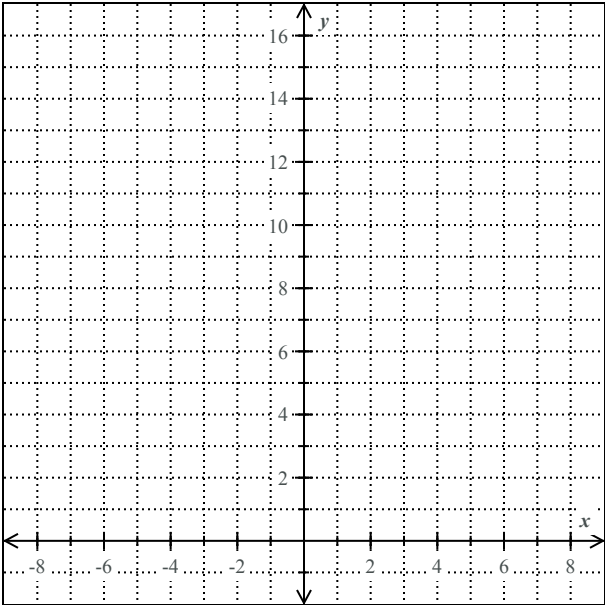
Find $h(x)$ for each x -value in the table.

x	$h(x)$
-2	_____
-1	_____
0	_____
1	_____
2	_____

Question 4 of 25

Graph the exponential function.

$$f(x)=\left(\frac{2}{5}\right)^x$$



Question 5 of 25

Hong deposited \$4000 into an account with 2.5% interest, compounded annually. Assuming that no withdrawals are made, how much will he have in the account after 7 years?

Do not round any intermediate computations, and round your answer to the nearest cent.

Question 6 of 25

A can of soda is placed inside a cooler. As the soda cools, its temperature $T(x)$ in degrees Celsius is given by the following function, where x is the number of minutes since the can was placed in the cooler.

$$T(x) = -8 + 28e^{-0.041x}$$

Find the initial temperature of the soda and its temperature after 15 minutes.
Round your answers to the nearest degree as necessary.

Initial temperature: _____ °C

Temperature after 15 minutes: _____ °C

Question 7 of 25

Rewrite each equation as requested.

(a) $\log_{\square} \square = \square$

(b) $\square^{\square} = \square$

(a) Rewrite as a logarithmic equation.

$$5^1 = 5$$

$$\log_{\square} \square = \square$$

(b) Rewrite as an exponential equation.

$$\log_2 \frac{1}{16} = -4$$

$$\square^{\square} = \square$$

Question 8 of 25

Rewrite each equation as requested.

(a) Rewrite as a logarithmic equation.

$$e^7 = y$$

Logarithmic equation: _____

(b) Rewrite as an exponential equation.

$$\ln x = 6$$

Exponential equation: _____

Question 9 of 25

Evaluate each expression.

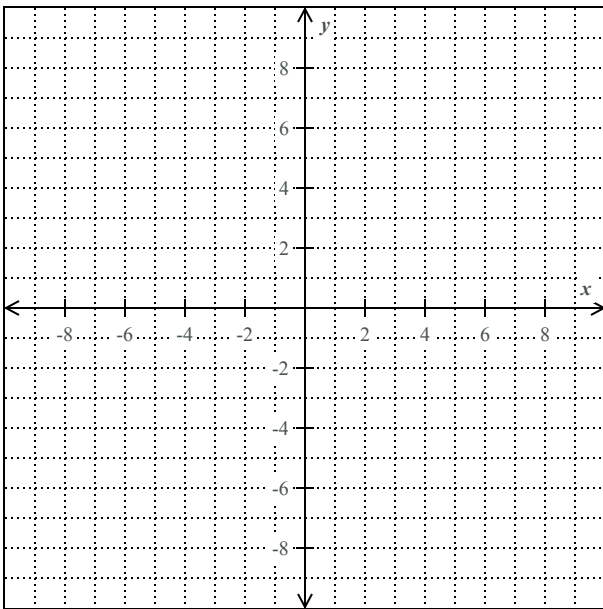
(a) $\log_9 1 = \square$

(b) $\log_7 49 = \square$

Question 10 of 25

- (a) Graph the logarithmic function $g(x) = -\log_4 x$.

To do this, plot two points on the graph of the function, and also draw the asymptote. Then, click on the graph-a-function button.



- (b) Give the domain and range of the function using interval notation.

Domain: _____

Range: _____

Question 11 of 25

Solve for x in the equation below.

Round your answer to the nearest hundredth.

Do not round any intermediate computations.

$$e^{x+4} = 3$$

$x =$ _____

Question 12 of 25

Use the properties of logarithms to expand the following expression.

$$\log \left(\frac{y^4}{x \sqrt[3]{z^2}} \right)$$

Each logarithm should involve only one variable and should not have any radicals or exponents.

You may assume that all variables are positive.

$$\log \left(\frac{y^4}{x \sqrt[3]{z^2}} \right) = \boxed{}$$

Question 13 of 25

Write the expression as a single logarithm.

$$4 \log_c (y - 2) - 5 \log_c (y + 5)$$

$\log$ _____ $\left(\text{_____} \right)$

Question 14 of 25

Use the change of base formula to compute $\log_{1/6} 5$.

Round your answer to the nearest thousandth.

Question 15 of 25

Solve for x .

$$\log_2(3x - 3) = 4$$

$$x = \boxed{}$$

Question 16 of 25

At the beginning of a study, a certain culture of bacteria has a population of 80. The population grows according to a *continuous exponential growth model*. After 12 days, there are 216 bacteria.

- (a) Let t be the time (in days) since the beginning of the study, and let y be the number of bacteria at time t .

Write a formula relating y to t .

Use exact expressions to fill in the missing parts of the formula.

Do not use approximations.

$$y = \boxed{} e^{\boxed{} t}$$

- (b) How many bacteria are there 22 days after the beginning of the study?

Do not round any intermediate computations, and round your answer to the nearest whole number.

$\boxed{}$ bacteria

Question 17 of 25

Rutherfordium-265 is a radioactive substance that decays according to the following function, where y_0 is the initial amount present, and y is the amount present at time t (in hours).

$$y = y_0 e^{-0.053t}$$

Find the half-life of Rutherfordium-265. Do not round any intermediate computations, and round your answer to the nearest tenth.

Question 18 of 25

Use a graphing calculator to solve the system of equations.

$$y = -3x - 6$$

$$y = 4x + 5$$

Round to the nearest hundredth.

Question 19 of 25

Use substitution to solve the system.

$$\begin{cases} 4x - 5y = 23 \\ 3x - 9 = y \end{cases}$$

$x =$	
$y =$	

Question 20 of 25

Solve the following system of equations.

$$-9x - 5y = 5$$

$$-7x + 9y = -9$$

Question 21 of 25

One month Manuel rented 2 movies and 3 video games for a total of \$21. The next month he rented 4 movies and 8 video games for a total of \$53. Find the rental cost for each movie and each video game.

Question 22 of 25

Let $C = \begin{bmatrix} 6 & 7 \\ 2 & 1 \\ 5 & -4 \end{bmatrix}$. Find $-4C$.

Question 23 of 25

Let $A = \begin{bmatrix} 0 & -6 \\ 4 & -4 \end{bmatrix}$ and $D = \begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix}$.

Find $A - D$.

Question 24 of 25

Yolanda, Ivan, and Bill each own a plane and take their own flights. They each spend 1.1 hours flying on Monday, 1.6 hours flying on Tuesday, and 1.5 hours flying on Wednesday. Their average speeds (in $\frac{\text{km}}{\text{h}}$) for each day are given in the following table.

Average speed $\left(\frac{\text{km}}{\text{h}}\right)$			
	Yolanda	Ivan	Bill
Monday	400	420	440
Tuesday	620	590	570
Wednesday	530	480	520

The table can be represented as the following matrix.

$$A = \begin{bmatrix} 400 & 420 & 440 \\ 620 & 590 & 570 \\ 530 & 480 & 520 \end{bmatrix}$$

(a) Find a matrix B whose entries are the times (in hours) spent flying each day. Make sure the product BA is defined and its entries are the distances (in kilometers) each pilot flew overall.

$B =$ km

(b) Calculate the product BA .

$BA =$

(c) How far (in kilometers) did Ivan fly overall?

_____ km

Question 25 of 25

Consider the following system of linear equations.

$$\begin{cases} 4x & -32z = 4 \\ -3x + 8y & = 13 \\ 5x + 2y - 48z & = -5 \end{cases}$$

Solve the system by completing the steps below to produce a reduced row-echelon form. R_1 , R_2 , and R_3 denote the first, second, and third rows, respectively. The arrow notation ($\rightarrow$) means the expression/matrix on the left becomes the expression/matrix on the right once the row operations are performed.

(a) For each step below, enter the coefficient for the row operation. (The first matrix in Step 1 is the augmented matrix for the given system of equations.) For Step 6, also fill in the missing matrix entries.

Step 1:

$$\left[\begin{array}{ccc|c} 4 & 0 & -32 & 4 \\ -3 & 8 & 0 & 13 \\ 5 & 2 & -48 & -5 \end{array} \right] \xrightarrow{\boxed{} \cdot R_1 \rightarrow R_1} \left[\begin{array}{ccc|c} 1 & 0 & -8 & 1 \\ -3 & 8 & 0 & 13 \\ 5 & 2 & -48 & -5 \end{array} \right]$$

Step 2:

$$\left[\begin{array}{ccc|c} 1 & 0 & -8 & 1 \\ -3 & 8 & 0 & 13 \\ 5 & 2 & -48 & -5 \end{array} \right] \xrightarrow{\begin{array}{l} 3 \cdot R_1 + R_2 \rightarrow R_2 \\ \boxed{} \cdot R_1 + R_3 \rightarrow R_3 \end{array}} \left[\begin{array}{ccc|c} 1 & 0 & -8 & 1 \\ 0 & 8 & -24 & 16 \\ 0 & 2 & -8 & -10 \end{array} \right]$$

Step 3:

$$\left[\begin{array}{ccc|c} 1 & 0 & -8 & 1 \\ 0 & 8 & -24 & 16 \\ 0 & 2 & -8 & -10 \end{array} \right] \xrightarrow{\boxed{} \cdot R_2 \rightarrow R_2} \left[\begin{array}{ccc|c} 1 & 0 & -8 & 1 \\ 0 & 1 & -3 & 2 \\ 0 & 2 & -8 & -10 \end{array} \right]$$

Step 4:

$$\left[\begin{array}{ccc|c} 1 & 0 & -8 & 1 \\ 0 & 1 & -3 & 2 \\ 0 & 2 & -8 & -10 \end{array} \right] \xrightarrow{\boxed{} \cdot R_2 + R_3 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & -8 & 1 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & -2 & -14 \end{array} \right]$$

Step 5:

$$\left[\begin{array}{ccc|c} 1 & 0 & -8 & 1 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & -2 & -14 \end{array} \right] \xrightarrow{\boxed{} \cdot R_3 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & -8 & 1 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & 1 & 7 \end{array} \right]$$

Step 6:

$$\left[\begin{array}{ccc|c} 1 & 0 & -8 & 1 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & 1 & 7 \end{array} \right] \xrightarrow{\begin{array}{l} 8 \cdot R_3 + R_1 \rightarrow R_1 \\ \boxed{} \cdot R_3 + R_2 \rightarrow R_2 \end{array}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & \boxed{} \\ 0 & 1 & 0 & \boxed{} \\ 0 & 0 & 1 & \boxed{} \end{array} \right]$$

(b) Give the solution.

$x = \boxed{}$

$y = \boxed{}$

$z = \boxed{}$

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Question 1 of 25

$$g^{-1}(2) = -5$$

$$h^{-1}(x) = 7x - 3$$

$$(h^{-1} \circ h)(4) = 4$$

Question 2 of 25

(a) $f(x) = x - 4$

$$g(x) = x + 4$$

$$f(g(x)) = x$$

$$g(f(x)) = x$$

- ☒ f and g are inverses of each other
☐ f and g are *not* inverses of each other

(b) $f(x) = \frac{1}{3x}, x \neq 0$

$$g(x) = -\frac{1}{3x}, x \neq 0$$

$$f(g(x)) = -x$$

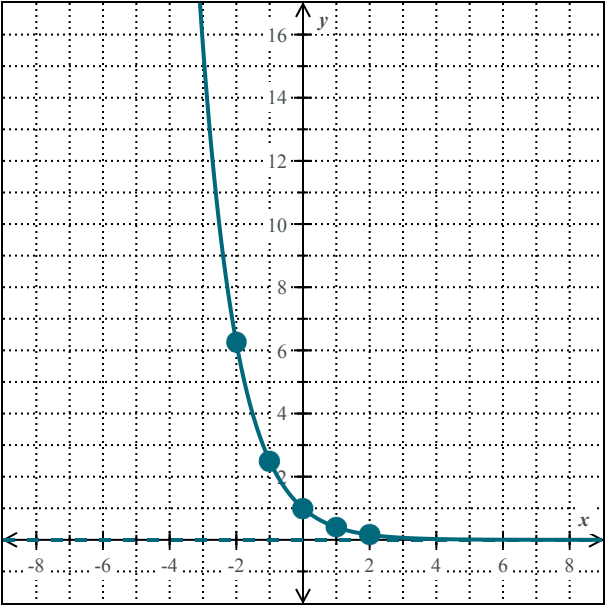
$$g(f(x)) = -x$$

- ☐ f and g are inverses of each other
☒ f and g are *not* inverses of each other

Question 3 of 25

x	$h(x)$
-2	100
-1	10
0	1
1	$\frac{1}{10}$
2	$\frac{1}{100}$

Question 4 of 25



Question 5 of 25

\$4754.74

Question 6 of 25

Initial temperature: $20\text{ }^{\circ}\text{C}$

Temperature after 15 minutes: $7\text{ }^{\circ}\text{C}$

Question 7 of 25

(a) $\log_5 5 = 1$

(b) $2^{-4} = \frac{1}{16}$

Question 8 of 25

(a) Logarithmic equation: $\ln y = 7$

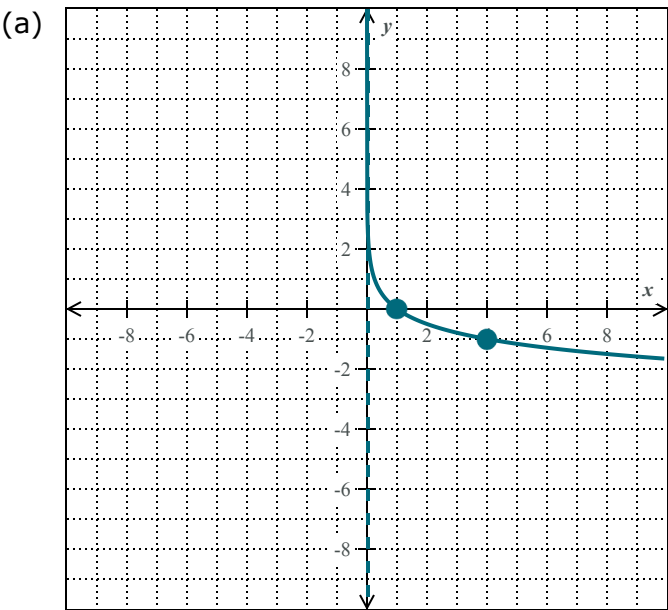
(b) Exponential equation: $e^6 = x$

Question 9 of 25

(a) $\log_9 1 = 0$

(b) $\log_7 49 = 2$

Question 10 of 25



(b)

Domain: $(0, \infty)$

Range: $(-\infty, \infty)$

Question 11 of 25

$x = -2.90$

Question 12 of 25

$$\log \left(\frac{y^4}{x \sqrt[3]{z^2}} \right) = 4 \log y - \log x - \frac{2}{3} \log z$$

Question 13 of 25

$$\log_c \left(\frac{(y-2)^4}{(y+5)^5} \right)$$

Question 14 of 25

−0.898

Question 15 of 25

$$x = \frac{19}{3}$$

Question 16 of 25

- (a) Let t be the time (in days) since the beginning of the study, and let y be the number of bacteria at time t .

Write a formula relating y to t .

Use exact expressions to fill in the missing parts of the formula.
Do not use approximations.

$$y = 80e^{\left(\frac{\ln 2.7}{12}\right)t}$$

- (b) How many bacteria are there 22 days after the beginning of the study?

Do not round any intermediate computations, and round your answer to the nearest whole number.

494 bacteria

Question 17 of 25

13.1 hours

Question 18 of 25

$$(x, y) = (-1.57, -1.29)$$

Question 19 of 25

$$x = 2$$

$$y = -3$$

Question 20 of 25

$$x = 0$$

$$y = -1$$

Question 21 of 25

Rental cost for each movie: \$2.25

Rental cost for each video game: \$5.50

Question 22 of 25

$$-4C = \begin{bmatrix} -24 & -28 \\ -8 & -4 \\ -20 & 16 \end{bmatrix}$$

Question 23 of 25

$$A - D = \begin{bmatrix} -1 & -10 \\ 5 & -1 \end{bmatrix}$$

Question 24 of 25

- (a) Find a matrix B whose entries are the times (in hours) spent flying each day. Make sure the product BA is defined and its entries are the distances (in kilometers) each pilot flew overall.

$$B = \begin{bmatrix} 1.1 & 1.6 & 1.5 \end{bmatrix} \text{ km}$$

- (b) Calculate the product BA .

$$BA = \begin{bmatrix} 2227 & 2126 & 2176 \end{bmatrix}$$

- (c) How far (in kilometers) did Ivan fly overall?

$$2126 \text{ km}$$

Question 25 of 25

(a) For each step below, enter the coefficient for the row operation. (The first matrix in Step 1 is the augmented matrix for the given system of equations.) For Step 6, also fill in the missing matrix entries.

Step 1:

$$\left[\begin{array}{ccc|c} 4 & 0 & -32 & 4 \\ -3 & 8 & 0 & 13 \\ 5 & 2 & -48 & -5 \end{array} \right] \xrightarrow{\frac{1}{4} \cdot R_1 \rightarrow R_1} \left[\begin{array}{ccc|c} 1 & 0 & -8 & 1 \\ -3 & 8 & 0 & 13 \\ 5 & 2 & -48 & -5 \end{array} \right]$$

Step 2:

$$\left[\begin{array}{ccc|c} 1 & 0 & -8 & 1 \\ -3 & 8 & 0 & 13 \\ 5 & 2 & -48 & -5 \end{array} \right] \xrightarrow{\begin{array}{l} 3 \cdot R_1 + R_2 \rightarrow R_2 \\ -5 \cdot R_1 + R_3 \rightarrow R_3 \end{array}} \left[\begin{array}{ccc|c} 1 & 0 & -8 & 1 \\ 0 & 8 & -24 & 16 \\ 0 & 2 & -8 & -10 \end{array} \right]$$

Step 3:

$$\left[\begin{array}{ccc|c} 1 & 0 & -8 & 1 \\ 0 & 8 & -24 & 16 \\ 0 & 2 & -8 & -10 \end{array} \right] \xrightarrow{\frac{1}{8} \cdot R_2 \rightarrow R_2} \left[\begin{array}{ccc|c} 1 & 0 & -8 & 1 \\ 0 & 1 & -3 & 2 \\ 0 & 2 & -8 & -10 \end{array} \right]$$

Step 4:

$$\left[\begin{array}{ccc|c} 1 & 0 & -8 & 1 \\ 0 & 1 & -3 & 2 \\ 0 & 2 & -8 & -10 \end{array} \right] \xrightarrow{-2 \cdot R_2 + R_3 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & -8 & 1 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & -2 & -14 \end{array} \right]$$

Step 5:

$$\left[\begin{array}{ccc|c} 1 & 0 & -8 & 1 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & -2 & -14 \end{array} \right] \xrightarrow{\frac{-1}{2} \cdot R_2 \rightarrow R_2} \left[\begin{array}{ccc|c} 1 & 0 & -8 & 1 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & 1 & 7 \end{array} \right]$$

Step 6:

$$\left[\begin{array}{ccc|c} 1 & 0 & -8 & 1 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & 1 & 7 \end{array} \right] \xrightarrow{\begin{array}{l} 8 \cdot R_3 + R_1 \rightarrow R_1 \\ 3 \cdot R_3 + R_2 \rightarrow R_2 \end{array}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 57 \\ 0 & 1 & 0 & 23 \\ 0 & 0 & 1 & 7 \end{array} \right]$$

(b) Give the solution.

$$x = 57$$

$$y = 23$$

$$z = 7$$